

Group M

Project name: - STUDENT ATTENDANCE MANAGEMENT SYSTEM)

Project Overview:

This project is about a digital solution designed to automate the process of recording, managing, and tracking student attendance.

Sprint	Duration	Focus	Goal
Sprint 1	Weeks 1-2	Foundation	Gather requirements, create system architecture, database design, project structure, implement OOP classes (Admin, Teacher, Student), and set up authentication (Registration, Login, Forgot Password).
Sprint 2	Weeks 3-4	Core Features	Develop role-based dashboards, student management, teacher management, manual attendance, attendance viewing, and attendance history features.
Sprint 3	Weeks 5-6	Advanced Features & Release	Implement facial recognition attendance, analytics dashboard, attendance percentage calculation, absent student lists, report generation, testing, bug fixes, and deployment preparation.

Functional Requirement

1. User Authentication

- The system shall allow users to register (students, admin, teacher).
- The system shall allow users to log in.
- The system shall allow the admin teacher to reset forgotten passwords.
- The system shall authenticate users based on their roles.

2. Student Management

- The system shall store student information in the database.
- The system shall allow students to view only their own attendance records.

3. Teacher Management

- The system shall allow teachers to log in.
- The system shall allow teachers to manually mark attendance.
- The system shall allow teachers to view student attendance (Total number of absences and presence)
- The system allows teachers to filter by student with their student ID and name for attendance check

4. Face Recognition Attendance

- The system shall allow students to register their faces.
- The system shall train the facial recognition model.
- The system shall automatically identify students using the camera.
- The system shall automatically mark attendance after successful recognition.
- The system shall prevent duplicate attendance entries for the same day.

5. Attendance Management

- The system shall store attendance records digitally.
- The system shall allow users to view attendance history.
- The system shall track absent students.
- The system shall calculate attendance percentages.

6. Dashboard

- The system shall display attendance statistics.
- The system shall display present and absent student counts.
- The system shall display attendance percentages.
- The system shall display a list of absent students.

7. Report Generation

- The system shall generate daily attendance reports.
- The system shall generate weekly attendance reports.
- The system shall generate monthly attendance reports.

8. Search and Filter

- The system shall allow searching for students by name.
- The system shall allow filtering attendance by date.
- The system shall allow filtering attendance by class.

9. Admin Functions

- The system shall allow the admin to view attendance records.
- The system shall allow the admin to generate reports.

Non-Functional Requirement

1. Performance

- The system should mark attendance within **3-5 seconds** after successful face recognition.
- Dashboard pages should load without noticeable delays.

2. Reliability

- The system should store attendance records accurately.
- Data should not be lost during normal operation.
- The database should remain consistent even if the application restarts.

3. Security

- Users must log in using valid credentials.
- Passwords should be securely stored (hashed instead of plain text if possible).
- Students can only view their own attendance records.
- Only authorized users can access teacher and admin functions.

4. Usability

- The interface should be simple and user-friendly.
- Navigation should be clear and consistent.
- Buttons, menus and dashboards should be easy to understand.

5. Maintainability

- The system should be modular and easy to modify.
- Functions should be reusable.
- The project should follow OOP principles.
- Proper comments and documentation should be provided.

6. Scalability

- The system should support adding more students and classes without major changes.
- The database structure should allow future expansion.

7. Availability

- The system should be available whenever authorized users need it.
- If deployed online, it should be accessible from different devices.

8. Readability

- The source code should use meaningful variable and function names.
- The code should follow a consistent style.
- The project should follow Pylint recommendations where possible.

9. Compatibility

- The system should work on common operating systems (Windows, macOS and Linux where applicable).
- If deployed online, it should work on modern web browsers.

10. Data Integrity

- Attendance records should not be duplicated.
- The system should validate data before storing it in the database.